

Project Title:

Retrieval Augmented Generation (RAG) System for Education Web Platform

Project Objective:

The objective of this project was to build an intelligent question-answering system for an education web platform.

The platform contains large volumes of unstructured data such as course content, FAQs, policies, study material,

and student guidance documents. Traditional keyword-based search was inefficient and often returned irrelevant results.

The RAG-based system enables accurate, context-aware, and real-time responses to student and learner queries.

Business Problem:

- Students frequently ask repetitive academic and administrative questions.
- Manual support and static FAQs do not scale well.
- LLMs alone tend to hallucinate without access to platform-specific knowledge.
- Need for trustworthy, grounded, and explainable answers using internal data.

Solution Overview:

We implemented a Retrieval Augmented Generation (RAG) architecture where user queries are first matched with the most relevant content from the platform's knowledge base, and then a Large Language Model (LLM) generates an answer strictly grounded in the retrieved information.

Data Used:

- Course descriptions and syllabus documents
- Learning content (PDFs, HTML pages)
- Admission and enrollment policies
- FAQs and support documents
- Student guidelines and project instructions

System Architecture:

1. Data Ingestion Layer – Content from the education platform is collected and stored.
2. Data Preprocessing – Documents are cleaned, chunked, and normalized.

3. Embedding Layer – Text chunks are converted into vector embeddings using transformer-based embedding models.
4. Vector Database – Embeddings are stored in a vector store for semantic similarity search.
5. Retriever – Retrieves top-k relevant chunks based on user query similarity.
6. LLM Generator – Generates final answers using retrieved context only.
7. Response Delivery – Answer returned to the web platform with citations.

Why RAG Approach:

- Reduces hallucinations by grounding responses in real data
- Keeps answers updated without model retraining
- Enables scalability across thousands of documents
- Improves trust and accuracy for educational users

Chunking Strategy:

- Fixed-size chunking with overlap to preserve context
- Metadata added (course name, topic, source type)
- Optimized chunk sizes to balance recall and performance

Vector Database & Retrieval:

- Vector similarity search using cosine similarity
- Implemented top-k retrieval with relevance score filtering
- Reranking logic applied for higher precision responses

Prompt Engineering:

- System prompt restricts the LLM to answer only from retrieved context
- If no relevant information is found, model returns a safe fallback response
- Clear instructions to maintain concise and student-friendly tone

Handling Hallucinations:

- Context-only answer generation enforced
- Confidence threshold on similarity scores
- Fallback responses for low-confidence matches

Evaluation Techniques:

- Precision and Recall on retrieved documents
- Human evaluation using expert-reviewed Q&A; pairs
- ROUGE score for similarity between generated and expected answers
- Error analysis on irrelevant retrievals and incomplete answers

Deployment:

- Backend API exposes the RAG pipeline
- Integrated with education web platform chatbot
- Stateless and scalable deployment using containerized services

Impact:

- Reduced support query resolution time significantly
- Improved student satisfaction with accurate answers
- Decreased manual intervention for academic support teams
- Enabled scalable AI-driven learning assistance

Interview Ready Summary:

“This project implemented a Retrieval Augmented Generation system for an education web platform to provide accurate,

context-aware answers using internal academic data. I worked on data preprocessing, vector embeddings, retrieval optimization,

prompt engineering, and evaluation to ensure high precision and minimal hallucinations. The solution improved student experience

and reduced operational support load.”